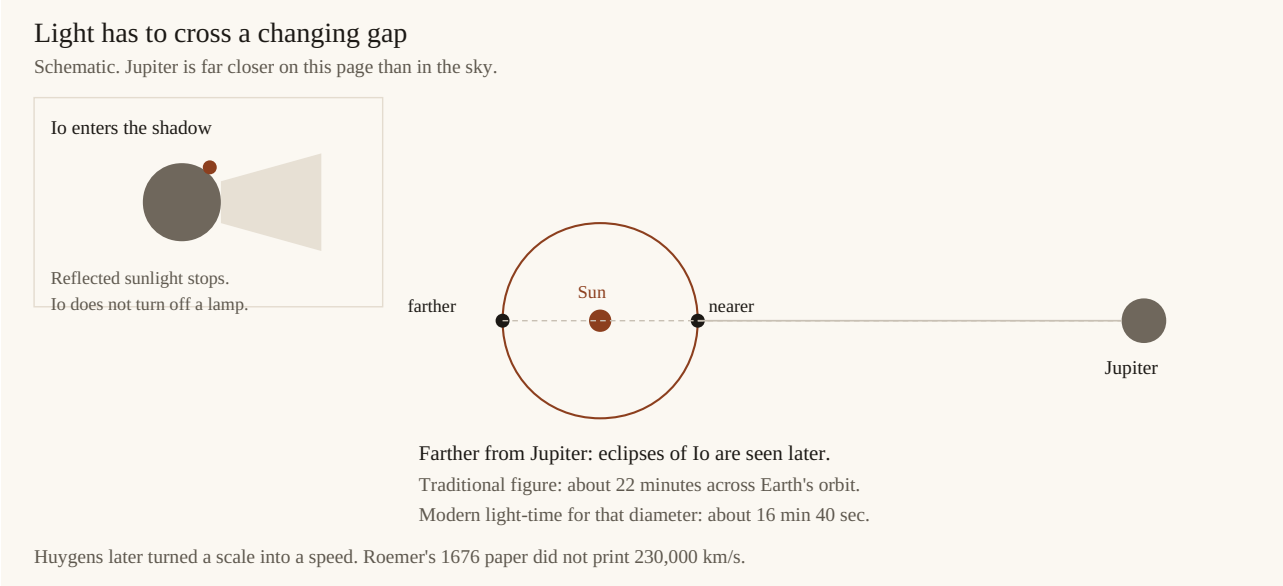
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Romer and Io: light needs time

Io goes around Jupiter in about 42.5 hours. We see it by sunlight reflected from its surface. When Io enters Jupiter's shadow, that sunlight is cut off. The moon has not switched off a lamp of its own. The disappearances and returns are eclipses, and they can be timed. The names and the order of the four moons are the previous lesson.

Giovanni Cassini had been building tables of those times. Ole Romer, working with that program in Paris, argued in 1676 that the tables made sense if light travels at a finite speed. Earth moves around the Sun once a year. When Earth is traveling away from Jupiter, each eclipse is seen a little later, because the light has a longer trip. When Earth is traveling toward Jupiter, the events come early. The extra minutes are the light-time of the extra distance.

Jupiter moves too, and Io's orbit is not a perfect clock. Many nights are required before the leftover can be blamed on light rather than on the tables. The traditional round number is about 22 minutes for light to cross the diameter of Earth's orbit. The modern light-time for that same diameter is about 16 minutes 40 seconds.



Io's shadow eclipse, and Earth near and far from Jupiter

A schematic, drawn for this lesson. Jupiter is much farther, and it moves. The inset is the local event: reflected light cut off by the shadow.

Romer did not publish a speed of 230,000 kilometers a second. Christiaan Huygens, in the Treatise on Light of 1690, converted an orbital scale into a speed that comes out near that figure in modern units. The defined speed of light is now 299,792.458 kilometers a second. A conversion is not a sentence Romer printed.

You also cannot pick one opposition and one conjunction and call them a clean pair. Near conjunction, Jupiter sits in the Sun's glare and Io's eclipses cannot be watched. The argument is a stack of dates. The drawing shows the idea, not the observing calendar.

The illustrated lesson is download the PDF.

Sources

- Paris Observatory, "340 years ago, Romer showed that light has a finite speed": https://observatoiredeparis.psl.eu/340-years-ago-romer-showed.html
- Observatoire de Paris, CLEA / CLIC notes on Romer: https://clic.obspm.fr/

Key facts

- Io's eclipses are the cutting off of reflected sunlight in Jupiter's shadow, on a clock of about 42.5 hours. Io does not emit the light.
- Romer in 1676 explained early and late eclipses by a finite speed of light as Earth moved nearer or farther.
- The traditional 22 minutes across Earth's orbit compares with a modern 16 minutes 40 seconds. Jupiter's motion and Io's orbit also affect the times.
- Huygens, not Romer, produced the speed later quoted near 230,000 km/s. The defined value is 299,792.458 km/s.